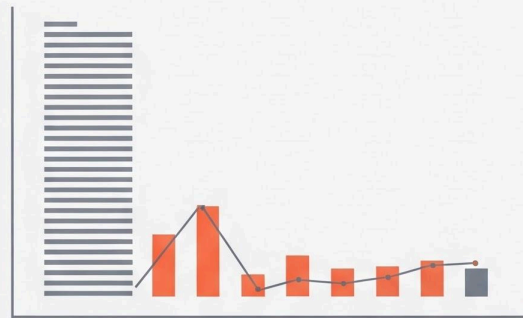


# IGN Reviews

## IGN Analytics



### Data Analytics Project: IGN Game Reviews Analysis

#### Dataset Used: ign\_csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

**Project Goal:** To analyze game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

## Project Coverage

| Category           | Skills Covered                                                                                    |
|--------------------|---------------------------------------------------------------------------------------------------|
| Data Preprocessing | Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal                 |
| Pandas Operations  | Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion) |
| NumPy Calculations | Array conversion, Yearly score growth calculation (np.diff)                                       |
| Visualization      | Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots                      |

### Scenario 1: Basic Data Loading & Cleaning (🟢 Easy)

#### Description:

This scenario is about loading and cleaning a dataset using Pandas. First, we read the data from a CSV file. Then we look at the data to understand it better. After that, we remove unwanted columns and check for missing values. Finally, we fix missing data and correct the data types so the dataset is ready for analysis.

-

| Task | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | Load the <code>ign.csv</code> dataset using Pandas.                                                                                                            |
| 2.   | Display the first 5 rows ( <code>head()</code> ), last 5 rows ( <code>tail()</code> ), and the shape of the dataset.                                           |
| 3.   | Remove the unnecessary column: " <code>Unnamed: 0</code> ".                                                                                                    |
| 4.   | Check for missing values in <code>score</code> , <code>genre</code> , and <code>platform</code> .                                                              |
| 5.   | Handle missing values: Fill the numeric column <code>score</code> with the mean, and the categorical column <code>genre</code> with the mode.                  |
| 6.   | Ensure correct data types: Convert <code>score</code> → float and <code>release_year</code> , <code>release_month</code> , <code>release_day</code> → integer. |

### Results of scenirio1:

```
   Unnamed: 0 score_phrase ... release_month release_day
0           0      Amazing ...           9         12
1           1      Amazing ...           9         12
2           2        Great ...           9         12
3           3        Great ...           9         11
4           4        Great ...           9         11

[5 rows x 11 columns]
   Unnamed: 0 score_phrase ... release_month release_day
18620      18620      Good ...           6         29
18621      18621    Amazing ...           6         29
18622      18622    Mediocre ...           6         28
18623      18623 Masterpiece ...           6         28
18624      18624 Masterpiece ...           6         28

[5 rows x 11 columns]
(18625, 11)
score           0
genre          36
platform       0
dtype: int64
score_phrase    str
title           str
url            str
platform       str
score          float64
genre          str
editors_choice str
release_year   int64
release_month  int64
release_day    int64
dtype: object
Yearly score growth: [-2.33825503  0.4045965  0.364337  0.0528054 -0.2898953
```

### Conclusion:

In this task, we successfully cleaned the dataset. We removed unnecessary columns and handled missing values properly. We also made sure all columns have the correct data types. Now the dataset is clean and ready to use for further analysis or visualization.

## Scenario 2: Line Graph (Score Trend) + Save (🟢 Easy)

### Description:

In this scenario, we analyze how game scores change over time. First, we group the data by release year. Then we calculate the average score for each year. After that, we convert the data into NumPy arrays for plotting. Finally, we create a line graph to show the trend and save the graph as an image file.

| Task | Description                                                                                     |
|------|-------------------------------------------------------------------------------------------------|
| 1.   | Group data by <code>release_year</code> .                                                       |
| 2.   | Calculate the average score per year using Pandas.                                              |
| 3.   | Convert the resulting average scores and years into NumPy arrays.                               |
| 4.   | Plot a line graph with <code>release_year</code> on the X-axis and average score on the Y-axis. |
| 5.   | Add the title: "Average Game Score Over Years" and proper axis labels.                          |
| 6.   | Save the graph: <code>plt.savefig("avg_score_trend.png")</code> .                               |

### Graph :



The graph is a line chart showing the average game score for each year. The X-axis shows the release year, and the Y-axis shows the average score. The line helps us see how scores go up or down over time.

### Conclusion:

This task shows the trend of game scores over the years. We can see that scores change over time, sometimes going up and sometimes going down. The graph makes it easy to understand these changes. Saving the graph allows us to use it later for reports or presentations.

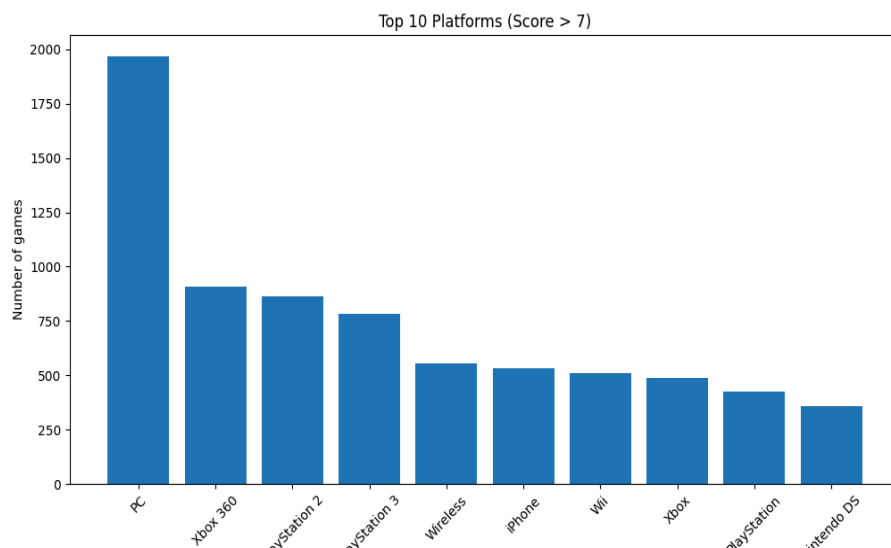
### Scenario 3: Filtering + Bar Chart + Save (🟡 Medium)

#### Description:

In this scenario, we work with a dataset of video games. First, we select only the games that have a score greater than 7. These are considered high-rated games. Then, we count how many high-rated games are available on each platform. After that, we pick the top 10 platforms with the highest counts. Finally, we create a bar chart to show this data clearly and save the chart as an image file.

| Task | Description                                                                                 |
|------|---------------------------------------------------------------------------------------------|
| 1.   | Filter the dataset to include only games where <code>score &gt; 7</code> .                  |
| 2.   | Count the number of high-rated games per platform.                                          |
| 3.   | Select the top 10 platforms using Pandas.                                                   |
| 4.   | Convert the platform counts into NumPy arrays.                                              |
| 5.   | Plot a bar chart with <code>platform</code> on the X-axis and count of games on the Y-axis. |
| 6.   | Rotate x-axis labels for readability.                                                       |
| 7.   | Save the graph: <code>plt.savefig("top_platforms_bar.png")</code> .                         |

#### Graph:



The bar chart shows different platforms on the X-axis and the number of high-rated games on the Y-axis. Platforms like PC, Xbox 360, and PlayStation have higher numbers compared to others.

#### Conclusion:

This analysis helps us understand which platforms have more high-rated games. PC has the highest number of good-rated games. Other platforms like Xbox 360 and PlayStation also perform well. The bar chart makes it easy to compare platforms visually. Saving the graph allows us to use it later for reports or presentations.

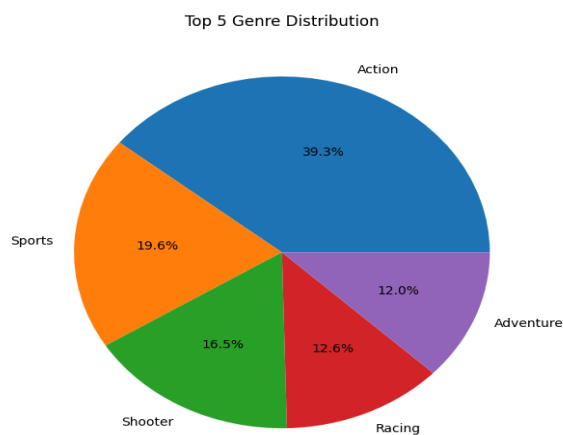
#### Scenario 4: Aggregation + Pie Chart + Save (🟡 Medium)

##### Description:

In this scenario, we analyze the number of games in each genre. First, we count how many games belong to each genre. Then, we select the top 5 genres with the highest number of games. After that, we prepare the data for a pie chart using genre names as labels and counts as values. Finally, we create the pie chart and save it as an image file.

| Task | Description                                                                           |
|------|---------------------------------------------------------------------------------------|
| 1.   | Count the total number of games per <code>genre</code> .                              |
| 2.   | Select the top 5 genres using Pandas.                                                 |
| 3.   | Prepare labels (genres) and values (counts) for the pie chart.                        |
| 4.   | Plot a pie chart using <code>genre</code> as labels and <code>count</code> as values. |
| 5.   | Add percentage labels using <code>autopct</code> (e.g., '%.1f%%').                    |
| 6.   | Save the graph: <code>plt.savefig("genre_distribution.png")</code> .                  |

##### Graph:



1. The pie chart shows the distribution of the top 5 genres.
2. The biggest portion is **Action**, which has the highest number of games.
3. Other genres like **Sports, Shooter, Racing, and Adventure** have smaller shares.
4. Each section shows the percentage of games in that genre.

### Conclusion:

This analysis shows which genres are most popular. Action games have the largest share among all genres. Sports and Shooter also have a good number of games. Racing and Adventure have fewer games compared to others. The pie chart helps us easily understand the distribution of game genres.

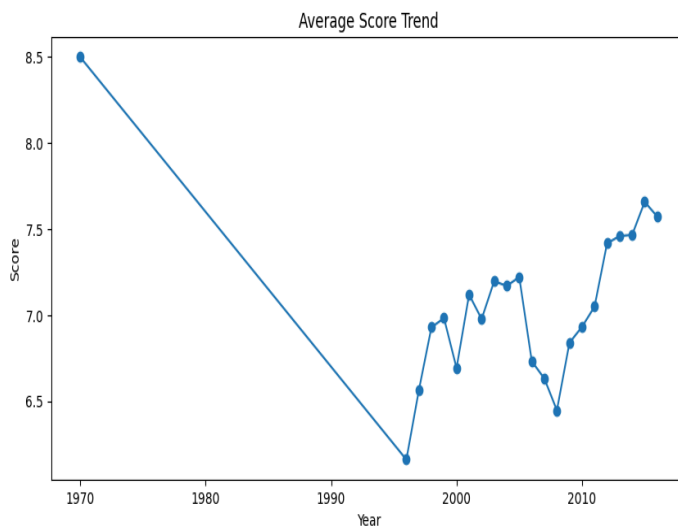
### **Scenario 5: Advanced Analysis + Multiple Graphs (🔴 Hard)**

#### Description:

In this scenario, we do advanced analysis on the game dataset. First, we create a new column called `score_category` based on score values. Then, we convert the `editors_choice` column into numeric form (1 and 0). After that, we calculate how scores change each year using NumPy. We create different graphs like a line chart, stacked bar chart, and histogram. Finally, we save all graphs for future use.

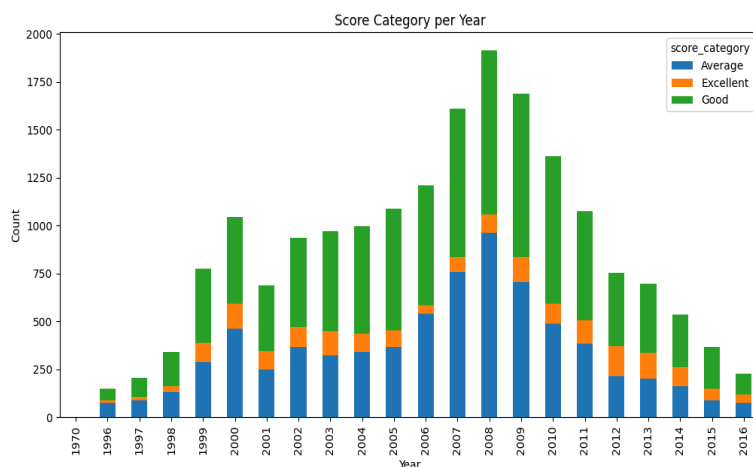
| Task | Description                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.   | Create a new column <code>score_category</code> : "Excellent" for score $\geq 9$ , "Good" for $7 \leq \text{score} < 9$ , and "Average" for score $< 7$ . |
| 2.   | Convert the <code>editors_choice</code> column: 'Y' $\rightarrow$ 1, 'N' $\rightarrow$ 0.                                                                 |
| 3.   | Use NumPy's <code>np.diff()</code> to calculate the <b>yearly score growth</b> based on the average yearly scores from Scenario 2.                        |
| 4.   | Plot a <b>Line Graph</b> showing the trend of average score per <code>release_year</code> .                                                               |
| 5.   | Plot a <b>Stacked Bar Chart</b> showing the count of <code>score_category</code> per <code>release_year</code> .                                          |
| 6.   | Plot a <b>Histogram</b> showing the distribution of <code>score</code> .                                                                                  |
| 7.   | Save the graphs:<br><code>plt.savefig("score_trend.png"), plt.savefig("score_category_stacked.png"), plt.savefig("score_distribution.png").</code>        |
| 8.   | Identify: Which years had the highest scores, whether high scores increased over time, and if <code>editors_choice</code> correlates with high scores.    |

### **Graph :**



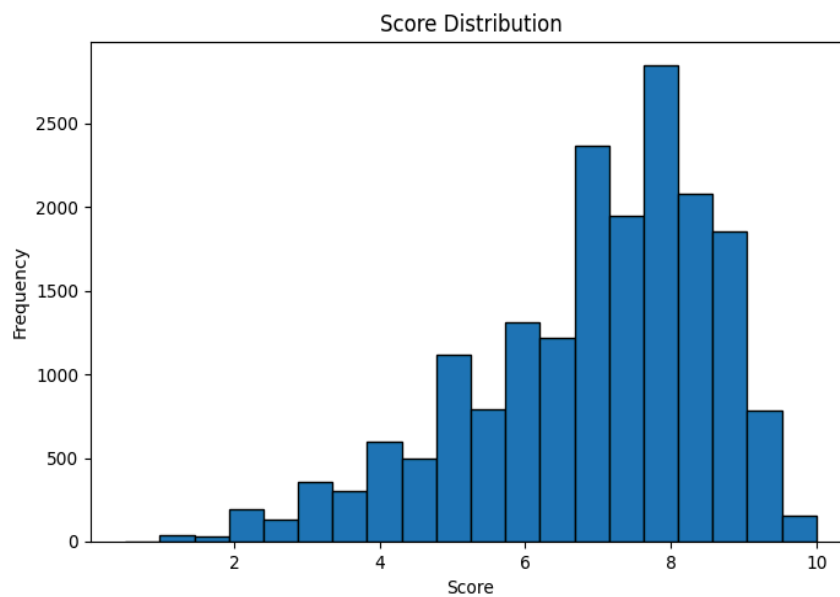
### **Line Graph (Average Score Trend):**

This graph shows how the average game score changes over the years. It helps us see if scores are improving or not.



### **Stacked Bar Chart (Score Category per Year):**

This graph shows how many games are Excellent, Good, and Average each year. It helps compare categories easily



#### **Histogram (Score Distribution):**

This graph shows how scores are spread. Most games have scores between 6 and 8.

#### **Results of scenario 5 :**

```
Yearly score growth: [-2.33825503  0.4045965  0.364337  0.0528054 -0.28989535  0.42690567
-0.14196855  0.21971997 -0.02753348  0.05171433 -0.48956223 -0.10031765
-0.18440037  0.3922213  0.09250116  0.11745752  0.36800005  0.04216181
0.00710926  0.19185429 -0.08600073]
Year with highest average score: 1970
No clear upward trend in scores
Correlation between editors choice and score: 0.5406719749441955
Strong positive correlation
```

#### **Conclusion:**

The analysis shows that some years have higher average scores than others. Around later years, scores slightly improve. Most games fall into the Good category. Very few games are rated Excellent. Also, games with editor's choice are usually high scoring, so there is a strong connection.

### **Final Conclusion:**

Across all scenarios, we cleaned, filtered, and analyzed the dataset step by step. We created different types of graphs like bar charts, pie charts, line graphs, and histograms. These visualizations helped us understand the data better. We also learned how to find trends and patterns in the dataset.

### **What you have learned:**

In this project, we worked with data step by step from basic to advanced level. First, we learned how to load and clean the data. Then, we filtered and grouped the data to get useful information. After that, we created different graphs like bar chart, pie chart, line graph, and histogram. These graphs helped us understand patterns and trends easily. Overall, this project improved our data analysis and visualization skills.